

Presentation Guide

Speaker notes and an explanation of every slide, in English and Nepali

हरेक slide को speaker note र बुझाइ, अङ्ग्रेजी र नेपालीमा

Jhala Nath Kafle (081MSPSE009) — DSDR_Presentation_Chaptered.pptx, 26 slides

How to use this guide. Each slide has four parts. **Say** is the speaker note: read it aloud, it is the same as in the deck's notes. **Understand** explains the slide in plain English, for you, not for reading aloud. **नेपालीमा** is the same explanation in Nepali. **If asked** gives a likely question and a short answer. The speaker notes total 724 words, about 6 minutes at a calm pace.

यो guide कसरी प्रयोग गर्ने। हरेक slide मा चार भाग छन्। **Say** भनेको बोल्ने कुरा हो: यसलाई ठूलो स्वरमा पढ्नुहोस्। **Understand** ले slide लाई सरल अङ्ग्रेजीमा बुझाउँछ; यो तपाईंको बुझाइका लागि हो, पढेर सुनाउनका लागि होइन। **नेपालीमा** भागमा त्यही बुझाइ नेपालीमा छ। **If asked** मा सोधिन सक्ने प्रश्न र छोटो उत्तर छ। प्राविधिक शब्दहरू (recloser, fuse, DG, CTI) अङ्ग्रेजीमै राखिएका छन्, किनभने defence मा तपाईंले तिनै शब्द प्रयोग गर्नुहुनेछ।

1. The whole story in one page

A distribution feeder is protected by a recloser and by fuses on its branches. Most faults on overhead lines are temporary. So the recloser is set to trip very fast first: the fault disappears, the recloser closes again, and the fuse is not lost. This is called fuse saving.

It only works if the recloser acts before the fuse starts to melt. Without a generator on the feeder this is easy, because the recloser and the fuse carry the same fault current.

When a distributed generator (DG) is connected, the fuse carries the current from the grid and from the DG together, but the recloser carries only its own share. The fuse now melts sooner, the recloser does not get faster, and the fuse can melt first. A recloser in the middle of the line even sees current flowing backwards.

The solution studied is the dual-setting directional recloser (DSDR). The mid-line recloser R2 gets two groups of settings: one for current coming from the grid (forward) and one for current coming from the DG (reverse). The direction of the current selects the group.

I built this on the IEEE 13-node feeder in DIGSILENT PowerFactory and checked 39 combinations of fault location and fault type. Without DG, 39 of 39 are coordinated. With the DG and ordinary settings, only 24. With the DSDR and larger fuses, 38 of 39.

The honest conclusion: the dual setting is necessary but not sufficient. It needs the fuse revision as well, and one case stays lost because the relay's dial cannot go any further.

पूरा कथा एक पृष्ठमा

Distribution feeder लाई recloser र शाखाहरूमा राखिएका fuse ले सुरक्षा गर्छन्। Overhead line मा हुने धेरैजसो fault अस्थायी हुन्छन्। त्यसैले recloser लाई पहिले धेरै छिटो trip गर्ने गरी मिलाइन्छ: fault हराउँछ, recloser फेरि बन्द हुन्छ, र fuse जोगिन्छ। यसलाई fuse saving भनिन्छ।

यो तब मात्र काम गर्छ जब fuse पगलन सुरु हुनुअघि नै recloser ले काम गर्छ। Feeder मा generator नहुँदा यो सजिलो हुन्छ, किनभने recloser र fuse दुवैमा उही fault current बग्छ।

Distributed generator (DG) जोडेपछि fuse मा grid र DG दुवैको current सँगै बग्छ, तर recloser मा आफ्नो भागको current मात्र बग्छ। अब fuse छिटो पगलन्छ, recloser भने छिटो हुँदैन, त्यसैले fuse पहिले पगलन सक्छ। Line को बीचमा रहेको recloser ले त उल्टो दिशाको current समेत देख्छ।

यसको समाधानका रूपमा dual-setting directional recloser (DSDR) अध्ययन गरिएको हो। बीचको recloser R2 लाई दुई समूहका setting दिइन्छ: एउटा grid बाट आउने current (forward) का लागि र अर्को DG बाट आउने current (reverse) का लागि। Current को दिशाले कुन समूह प्रयोग गर्ने भन्ने छान्छ।

मैले यो विधि DIGSILENT PowerFactory मा IEEE 13-node feeder मा बनाएँ र fault को स्थान र प्रकारका 39 वटा संयोजन जाँचें। DG नहुँदा 39 मध्ये 39 मा coordination कायम रहन्छ। DG जोडेर साधारण setting राख्दा 24 मा मात्र। DSDR र ठूला fuse राख्दा 39 मध्ये 38 मा।

इमानदार निष्कर्ष: dual setting आवश्यक छ तर एकलै पर्याप्त छैन। यसलाई fuse को आकार पनि बदल्नुपर्छ, र एउटा अवस्था अझै बिग्रिएको रहन्छ, किनभने relay को dial त्योभन्दा बढाउन मिल्दैन।

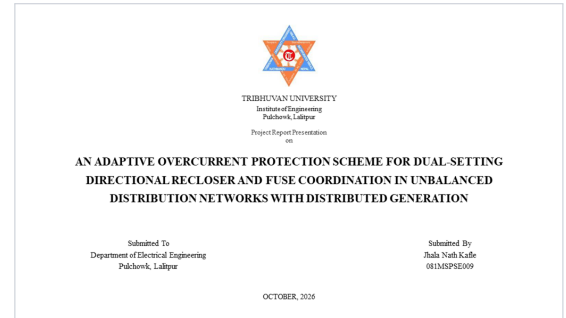
2. Key terms

Term	Meaning	नेपालीमा
Recloser	A breaker with a relay that trips, waits, and closes again by itself.	आफैं trip गर्ने, केही समय पर्खने र फेरि आफैं बन्द हुने breaker।
Fuse	A wire that melts when too much current flows; it must be replaced by hand.	धेरै current बग्दा पग्लने तार; पल्लिएपछि मान्छेले नै बदल्नुपर्छ।
Fuse saving	Letting the recloser clear a temporary fault before the fuse melts.	Fuse पग्लनुअघि नै recloser ले अस्थायी fault हटाउने तरिका।
DG	Distributed generator: a generator connected inside the distribution feeder.	Distribution feeder भित्रै जोडिएको generator।
Coordination	The devices operate in the right order.	उपकरणहरूले सही क्रममा काम गर्नु; तालमेल।
CTI	Coordination time interval = fuse melting time minus recloser fast time. Must be positive.	Fuse पग्लने समय र recloser को छिटो trip समयको फरक। यो धनात्मक हुनुपर्छ।
Pickup current	The current above which the relay starts to operate.	जति current नाघेपछि relay ले काम सुरु गर्छ, त्यो मान।
Time dial (TDS, TMS)	A setting that makes the whole relay curve slower or faster.	Relay को पूरै curve लाई ढिलो वा छिटो बनाउने setting।
Fast and delayed curve	The recloser's two curves: fast for temporary faults, delayed for permanent ones.	Recloser का दुई curve: अस्थायी fault का लागि छिटो, स्थायी fault का लागि ढिलो।
MMT and TCT	Minimum melting time and total clearing time of a fuse.	Fuse पग्लन सुरु हुने न्यूनतम समय र fault पूरै हटाउन लाग्ने कुल समय।
Cell	One fault location with one fault type. There are 39.	एउटा स्थान र एउटा प्रकारको fault को जोडी। जम्मा 39 वटा छन्।
Forward and reverse	Current from the grid side, and current from the DG side.	Grid तिरबाट आउने current र DG तिरबाट आउने current।
EMT	A simulation that follows the current wave in time.	Current को तरङ्गलाई समयसँगै पछ्याउने simulation।
Penetration	How large the DG is, as a percentage of its full size.	DG कति ठूलो छ भन्ने, पूरा क्षमताको प्रतिशतमा।

3. Slide by slide

SLIDE 1

Title



Say

Good afternoon, respected teachers and friends. I am Jhala Nath Kafle, roll number 081 MSPSE 009. My project is on dual-setting directional recloser and fuse coordination in distribution networks with distributed generation.

Understand

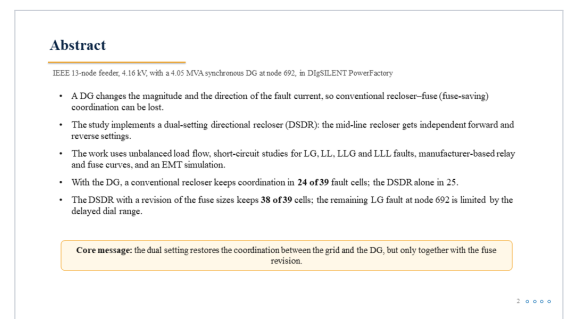
The title names three things: the device (a recloser with two settings chosen by the direction of the current), what it must work with (the fuses), and the situation (a feeder with a generator connected). In one line: how to keep the recloser and the fuses working in the right order after a generator is added.

नेपालीमा

शीर्षकले तीन कुरा भन्छ: उपकरण (current को दिशा अनुसार छानिने दुई setting भएको recloser), जोसँग मिलेर काम गर्नुपर्छ (fuse), र अवस्था (generator जोडिएको feeder)। एक वाक्यमा: generator थपेपछि पनि recloser र fuse ले सही क्रममा काम गरिरहने कसरी बनाउने।

SLIDE 2

Abstract



Say

A DG can break recloser-fuse coordination. I implemented a dual-setting directional recloser on the IEEE 13-node feeder. With the DG, coordination holds in 24 of 39 cells. With the DSDR and revised fuses, it holds in 38 of 39.

Understand

This slide is the whole project in four lines: the problem, the method, the tools, and the result. The two numbers to remember are 24 and 38. The yellow box is the message you want the examiners to keep: the dual setting works, but only together with larger fuses.

नेपालीमा

यो slide मा पूरै project चार पङ्क्तिमा छ: समस्या, विधि, प्रयोग गरिएका साधन र नतिजा। सम्झनुपर्ने दुई सङ्ख्या 24 र 38 हुन्। पहुँलो बाकसमा भएको कुरा नै परीक्षकले सम्झ्नु भन्ने मुख्य सन्देश हो: dual setting ले काम गर्छ, तर fuse को आकार बढाएपछि मात्र।

If asked

What is a cell? One fault location with one fault type. Twelve locations and four fault types give 39 cells, because not every node has all three phases.

SLIDE 3

Presentation Roadmap

Presentation Roadmap

Chapter I	Introduction, problem statement, objectives and scope
Chapter II	Literature review, research gap and positioning of the project
Chapter III	Methodology: test system, flowchart, formulation and protection settings
Chapter IV	Results and discussion: coordination without DG and with DG; DG penetration; the DSDR: EMT check
Chapter V	Conclusion, recommendation and future scope
References	Method, test feeder and device sources

Say

I will go through the introduction, the literature, the methodology, the results and the conclusion.

Understand

Just the order of the talk. Say it in one breath and move on.

नेपालीमा

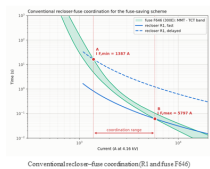
प्रस्तुतिको क्रम मात्र हो। एकै सासमा भनेर अगाडि बढ्नुहोस्।

SLIDE 4

Chapter I: Background and Problem

Chapter I: Background and Problem

- Fuse saving: the recloser trips on its fast curve before the lateral fuse melts; its delayed curve lets the fuse clear a permanent fault.
- Coordination holds only while the recloser and the fuse carry the **same current**.
- With a DG the fuse carries grid + DG current; the recloser only the grid share.
- A mid-line recloser can see current in the **reverse direction**.
- The fuse can melt before the fast trip: fuse saving is lost.



Problem focus: can independent forward and reverse settings of the mid-line recloser restore the recloser-fuse coordination when a DG is connected?

Say

The recloser trips fast to save the fuse on a temporary fault. That works only while both carry the same current. With a DG, the fuse carries more current than the recloser, and it can melt first.

Understand

Look at the graph. The solid blue line is the recloser's fast curve, the green band is the fuse, and the dashed line is the recloser's delayed curve. For fuse saving the fast curve must be below the fuse and the delayed curve above it. This is true only between points A and B. A DG pushes the fuse current to the right, outside that range, while the recloser current stays where it was.

नेपालीमा

ग्राफ हेर्नुहोस्। निलो सिधा रेखा recloser को छिटो curve हो, हरियो पट्टी fuse हो, र थोप्ले रेखा recloser को ढिलो curve हो। Fuse saving का लागि छिटो curve fuse भन्दा तल र ढिलो curve fuse भन्दा माथि हुनुपर्छ। यो A र B बिन्दुको बीचमा मात्र सत्य हुन्छ। DG ले fuse को current लाई दायाँतिर, त्यो दायराभन्दा बाहिर धकेल्छ, तर recloser को current भने जहाँको त्यहीँ रहन्छ।

If asked

Why does the recloser not see the DG current? The DG is connected beyond the recloser, closer to the fault. Its current goes straight into the fault through the fuse and never passes the recloser.

SLIDE 5

Chapter I: Objectives and Study Scope

Chapter I: Objectives and Study Scope

General Objective

To build and validate a DigSILENT PowerFactory implementation of the dual-setting directional recloser (DSDR) coordination method on the IEEE 13-node feeder, with and without distributed generation.

Specific Objectives

1. To build and verify the IEEE 13-node feeder with its recloser and fuse protection.
2. To perform unbalanced load-flow and short-circuit studies for LG, LL, LLG and LLL faults.
3. To calculate the recloser pickups, time dials and fuse coefficients.
4. To establish the coordination without DG, and with a 4.05 MVA synchronous DG at node 692.
5. To design the DSDR settings of R2 and classify the resulting coordination.
6. To verify the reclosing sequence in the time domain (EMT simulation).
7. To study the effect of the DG penetration level from 0 to 100 %.

Scope: IEEE 13-node feeder; 12 fault locations × 4 fault types = 39 cells; one DG location (692); zero-margin criterion CTI > 0.

Say

The general objective is to build and validate the DSDR method on the IEEE 13-node feeder in PowerFactory. There are seven specific objectives, from building the model to the DG penetration study.

Understand

The general objective is the one big aim. The seven specific objectives are the steps taken to reach it, in the order the work was done: build the model, run the studies, calculate the settings, check without DG and with DG, apply the dual setting, verify in time, and vary the DG size. Do not read all seven aloud.

नेपालीमा

General objective भनेको एउटा मुख्य लक्ष्य हो। सात वटा specific objective त्यो लक्ष्यमा पुग्न चालिएका कदम हुन्, काम गरिएका क्रममा: model बनाउने, अध्ययन चलाउने, setting निकाल्ने, DG बिना र DG सहित जाँच्ने, dual setting लगाउने, समयमा प्रमाणित गर्ने, र DG को आकार फेरि हेर्ने। सातै वटा ठूलो स्वरमा नपढ्नुहोस्।

SLIDE 6

Chapter II: Literature Positioning and Gap

Chapter II: Literature Positioning and Gap

Literature stream	Representative sources	Relevance and remaining gap
Recloser-fuse coordination with DG	Naim et al. [1]	Classifies when coordination holds or is lost with DG; fixed, single settings.
Dual-setting directional recloser	Yousaf et al. [1]	The method implemented here; R2 settings and fuse sizes are not fully specified.
Relay characteristics	Benmouyal et al. [2]; GE Multilin [4]	Inverse-time equations; the method uses a generic curve; real relays have fixed curves and dial ranges.
Test feeder and benchmark	Kersting [3]; Kersting & Shirek [7]; Yousaf et al. [6]	IEEE 13-node data and short-circuit 'benchmark' used to validate the model.

Gap addressed: an independent PowerFactory implementation of the DSDR method with manufacturer-based curves, real dial ranges and standard fuse sizes.

Say

The literature covers coordination with DG, the DSDR method, relay curves and the test feeder. The gap is an independent implementation with real relay curves, dial ranges and fuse sizes.

Understand

Each row is a group of earlier work and what it leaves open. The key row is the second: the DSDR method comes from the reference paper, but that paper does not give all the settings and fuse sizes. The gap this project fills is to build the method independently, with the curves and limits of real devices.

नेपालीमा

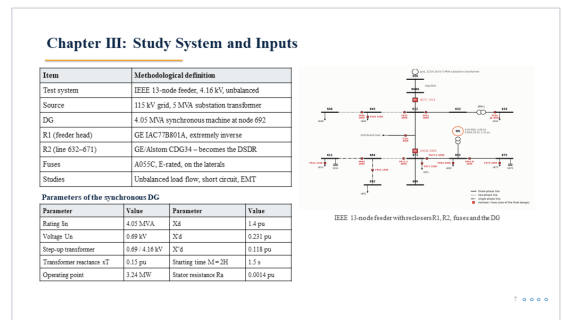
हरेक पङ्क्तिमा पहिलेका कामको एउटा समूह र त्यसले बाँकी छोडेको कुरा छ। मुख्य पङ्क्ति दोस्रो हो: DSDR विधि reference paper बाट आएको हो, तर त्यो paper मा सबै setting र fuse को आकार दिइएको छैन। यो project ले पूरा गर्ने खाली ठाउँ भनेको वास्तविक उपकरणका curve र सीमासहित यो विधिलाई स्वतन्त्र रूपमा बनाउनु हो।

If asked

Is your work only a copy of the paper? No. It is an independent implementation. I used the manufacturers' curves and real dial ranges, validated the model against the IEEE benchmark, and found where the method reaches the limits of the devices.

SLIDE 7

Chapter III: Study System and Inputs



Say

This is a 4.16 kV feeder with a 4.05 MVA synchronous DG at node 692. R1 is at the feeder head, R2 is on line 632 to 671, and fuses protect the laterals.

Understand

The diagram shows where everything is. Power comes in at the top. R1 sits at the head of the feeder and sees every fault. R2 sits in the middle. The DG is at node 692, below R2. So for a fault above R2, the DG current flows backwards through R2. That is the reason R2 needs a reverse setting. The lower table gives the generator's data.

नेपालीमा

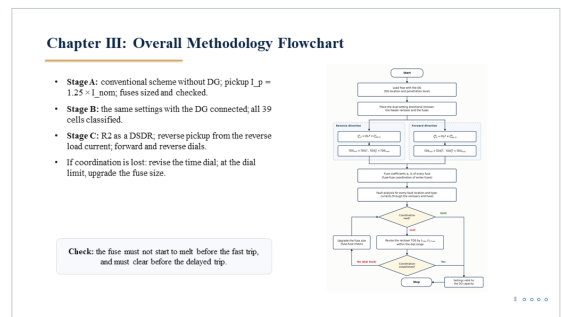
चित्रले कुन चिज कहाँ छ भनेर देखाउँछ। बिजुली माथिबाट आउँछ। R1 feeder को सुरुमा छ र हरेक fault देख्छ। R2 बीचमा छ। DG node 692 मा, R2 भन्दा तल छ। त्यसैले R2 भन्दा माथिको fault मा DG को current R2 बाट उल्टो दिशामा बग्छ। R2 लाई reverse setting चाहिनुको कारण यही हो। तलको तालिकामा generator का विवरण छन्।

If asked

Why node 692 for the DG? It is the location used in the reference method. It is downstream of R2, so it creates reverse current in R2 for faults upstream.

SLIDE 8

Chapter III: Overall Methodology Flowchart



Say

A load flow gives the branch currents, and each recloser's pickup is set at 1.25 times that current. R2 gets separate forward and reverse settings, for the grid and the DG. The fuses are sized, and then every fault is checked. If coordination fails, the time dial is adjusted first, and the fuse is made larger if necessary.

Understand

Read the flowchart from top to bottom. First find the normal current in each branch. The relay must not trip on normal current, so its pickup is set 25 per cent higher. For R2 this is done twice, once for each direction. Then test every fault. If the fuse would melt before the recloser trips, try to fix it with the dial. If the dial is already at its end, use a larger fuse and test again.

नेपालीमा

Flowchart लाई माथिबाट तल पढ्नुहोस्। पहिले हरेक शाखामा सामान्य अवस्थाको current निकालिन्छ। Relay ले सामान्य current मा trip गर्नु हुँदैन, त्यसैले pickup लाई 25 प्रतिशत बढी राखिन्छ। R2 का लागि यो दुई पटक गरिन्छ, हरेक दिशाका लागि एक-एक

पटक। त्यसपछि हरेक fault जाँचिन्छ। Recloser ले trip गर्नुअघि नै fuse पग्लने देखिए पहिले dial मिलाएर सुधार्ने प्रयास गरिन्छ। Dial पहिले नै अन्तिम सीमामा छ भने ठूलो fuse राखेर फेरि जाँचिन्छ।

If asked

Why 1.25? It is the overload factor of the method. It keeps the relay from tripping on normal load, with a 25 per cent allowance.

SLIDE 9

Chapter III: Mathematical Formulation of the Method

Chapter III: Mathematical Formulation of the Method

1 Recloser operating time (forward fw, reverse rv)

$$t_{r,f} = TDS_{r,f} \left[\frac{A}{(I_{r,f}/I_{r,n})^2 - 1} + B \right]$$

$$t_{r,r} = TDS_{r,r} \left[\frac{A}{(I_{r,r}/I_{r,n})^2 - 1} + B \right]$$

$A \in \{fw, rv\}$; F = fast, D = delayed; one curve, two time dial

2 Pickup and constraints

$$I_{r,n}^{pick} = OLF \times I_{r,n}^{rated}$$

$$I_{f,n}^{pick} = OLF \times I_{f,n}^{rated}$$

$$TDS_{min} < TDS_{r,f}, TDS_{r,r} < TDS_{max}$$

$$I_{f,n} < I_{r,f} < I_{f,max} \quad (OLF = 1.25)$$

3 Fuse characteristic and coefficient

$$\log t_{f,max} = a \log I_f + b_1 \quad (a = -1.8)$$

$$b_1 = \log \left[\frac{(I_{f,r} + I_{f,d})}{2} \right] - a \log I_{f,r} \quad (\text{single fuse})$$

$$b_1 = \log \left[I_{f,r} + \frac{(I_{f,d} - I_{f,r})}{2 + 1} \right] - a \log I_{f,r} \quad (z \text{ in series})$$

4 Coordination conditions (fuse saving)

$$CTI = t_{fmax}(I_{f,max}) - t_{r,f}(I_{f,max}) > 0 \quad \text{fast trip first}$$

$$t_{f,r}(I_{f,n}) < t_{r,d}(I_{f,n}) \quad \text{fuse clears first}$$

$$t_{f,r} \leq 0.75 \, t_{fmax,r+1} \quad \text{series fuses}$$

Say

These are the method's equations: the recloser curve, the pickup, the fuse line and the coordination conditions.

Understand

Four boxes. Box 1: the time a recloser takes depends on how many times the fault current is above the pickup; a bigger current gives a shorter time. Box 2: the pickup is 1.25 times the rated current. Box 3: the fuse curve is a straight line on log-log paper. Box 4: the two conditions that mean coordination holds. Point at box 4 if time is short.

नेपालीमा

चार वटा बाकस छन्। बाकस 1: recloser लाई लाग्ने समय fault current pickup भन्दा कति गुणा बढी छ भन्नेमा भर पर्छ; current जति ठूलो, समय त्यति छोटो। बाकस 2: pickup भनेको rated current को 1.25 गुणा हो। बाकस 3: log-log कागजमा fuse को curve सिधा रेखा हुन्छ। बाकस 4: coordination कायम छ भन्ने जनाउने दुई सर्त। समय कम भए बाकस 4 मात्र देखाउनुहोस्।

If asked

What does CTI greater than zero mean? The fuse starts to melt later than the recloser's fast trip. So the recloser acts first and the fuse is saved.

SLIDE 10

Chapter III: Formulation Used for the Operating Times

Chapter III: Formulation Used for the Operating Times

Method: $t = TDS \left[\frac{A}{(M-C)^2} + B \right]$ $\implies$ in general: $t = TDS \times g(M)$, $M = I_d/I_p$ $g(M)$ = curve of the specific relay

R1: GE IAC78801A, extremely inverse

$$g(M) = A + \frac{B}{M-C} + \frac{D}{(M-C)^2} + \frac{E}{(M-C)^3}$$

$A = 0.004$, $B = 0.6379$, $C = 0.62$, $D = 1.7872$, $E = 0.2461$

GE Multilin manual, $I_p = 720$ A, TDS 0.5 / 10

R2 (DSOR): GE/Alstom CDG34, extremely inverse

$$g(M) = t_0 \left(\frac{I_p}{M} \right)^n$$

$t_0 = \frac{\ln(M/M_0)}{\ln(M_0/M_1)}$

manufacturer table (PowerFactory library), $M = I_d/I_p$

fwd 1, 300/600 A, rev 150/300 A, TMS 0.1 / 1.0

- The method fixes the structure $t = TDS \times g(M)$, with $M = I_d/I_p$, the constants A, B and n describe only a generic curve.
- So $g(M)$ is taken from the manufacturer of each relay: the IAC extremely inverse equation for R1, the CDG34 table for R2.
- Check: R1 gives 0.096 / 1.926 s at 4219 A, against 0.097 / 1.932 s in the reference paper.

10

Say

I kept the method's structure and used each manufacturer's curve: the GE IAC equation for R1 and the CDG34 table for R2. R1 reproduces the paper's worked example.

Understand

The method's formula has constants A, B and n that describe a general curve. A real relay has its own fixed curve from its manufacturer. So the form 'time = dial times a function of current' is kept, and the function is taken from the real device. The proof that this is right: for 4219 amperes R1 gives 0.096 seconds, and the paper gives 0.097.

नेपालीमा

विधिको सूत्रमा A, B र n भन्ने स्थिराङ्क छन्, जसले एउटा सामान्य curve जनाउँछन्। वास्तविक relay को भने निर्माताले दिएको आफ्नै निश्चित curve हुन्छ। त्यसैले 'समय = dial गुणा current को function' भन्ने ढाँचा जस्ताको तस्तै राखिएको छ, र function चाहिँ वास्तविक उपकरणबाट लिइएको छ। यो ठीक छ भन्ने प्रमाण: 4219 ampere मा R1 ले 0.096 सेकेन्ड दिन्छ, र paper मा 0.097 छ।

If asked

Why did you change the paper's formula? I did not change the method. I replaced the generic curve with the curve of the actual relay, because a real relay cannot follow arbitrary constants. The pickups, dials and coordination rules are the method's.

SLIDE 11

Chapter III: Protection Settings

Chapter III: Protection Settings

Device	Characteristic	Pickup / plug (fast / delayed)	Dial (fast / delayed)
R1 (feeder load)	GE IAC extremely inverse	720 A	TDS 0.5 / 10
R2 forward (grid current)	CDG34	300 / 600 A	TMS 0.1 / 1.0
R2 reverse (DG current)	CDG34, CT 500/5	150 / 300 A	TMS 0.1 / 1.0

- Load current through R2: ~470 A without DG, ~252 A (reversed) with the DG → reverse pickup $1.25 \times 252 = 315$ A.
- A dial revision could not help: R1 and R2 are already at the ends of their dial ranges.
- Fuse revision: F632 → 500E; F633, F646, F-DL, F671-1 → 400E; F692-R → 250E.

TDS (IAC relay) and TMS (CDG34 relay) are the manufacturers' names for the same thing: the dial that scales the curve.

11

Say

R1 picks up at 720 amperes. R2 has a forward group of 300 and 600 amperes, and a new reverse group of 150 and 300. The dials were already at their limits, so the fuses were revised.

Understand

The table is the result of the design. R2 has two rows: forward for grid current, reverse for DG current. The reverse values are smaller because the DG is a smaller source than the grid. The first bullet is the key fact: without DG, 470 amperes flow one way through R2; with DG, 252 amperes flow the other way.

नेपालीमा

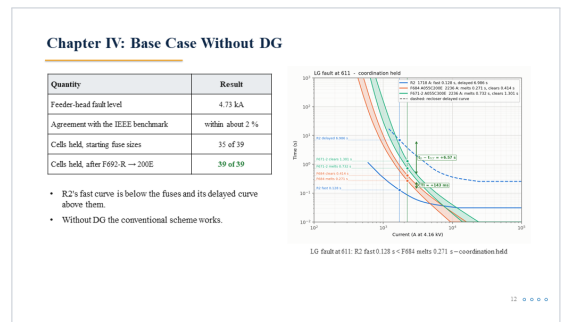
यो तालिका design को नतिजा हो। R2 का दुई पङ्क्ति छन्: grid को current का लागि forward र DG को current का लागि reverse। Reverse का मान साना छन्, किनभने DG grid भन्दा सानो स्रोत हो। पहिलो बुँदा नै मुख्य तथ्य हो: DG नहुँदा R2 बाट 470 ampere एक दिशामा बग्छ; DG हुँदा 252 ampere अर्को दिशामा बग्छ।

If asked

Why is R1's dial called TDS and R2's called TMS? They are the two manufacturers' names for the same thing: the setting that scales the curve.

SLIDE 12

Chapter IV: Base Case Without DG



Say

Without DG, 35 of 39 cells are coordinated, and all 39 after one fuse change.

Understand

This is the starting point: the feeder works properly before the DG is added. The plot shows one fault. R2 trips at 0.128 seconds and the fuse would melt at 0.271 seconds, so the recloser is first. The four cells lost at the start were a fuse-to-fuse problem, fixed by changing one fuse size.

नेपालीमा

यो सुरुको अवस्था हो: DG थप्नुअघि feeder ले राम्ररी काम गर्छ। ग्राफले एउटा fault देखाउँछ। R2 ले 0.128 सेकेन्डमा trip गर्छ र fuse 0.271 सेकेन्डमा मात्र पग्ल्थ्यो, त्यसैले recloser अगाडि छ। सुरुमा बिग्रिएका चार cell दुई fuse बीचको समस्या थिए, एउटा fuse को आकार बदलेर सुधारियो।

SLIDE 13

Chapter IV: Branch Currents and Fault Levels - Side by Side

Chapter IV: Branch Currents and Fault Levels - Side by Side

Branch	1 min (A) study	1 min (A) paper	1 min (A) IEEE	1 min (A) IEEE (%)	1 min (A) study	1 min (A) paper	1 min (A) IEEE (%)	Study vs IEEE	Paper vs IEEE
0000-002	587.7	587.7	1.07	1.24	4.80	4.78	1.41	-1.3%	-15.7%
002-003	8.0	8.1	0.76	0.72	4.32	4.30	4.70	-1.3%	-45.2%
002-004	140.1	140.3	0.73	0.93	3.41	3.42	4.23	-0.2%	-24.6%
002-071	470.0	478.2	0.88	1.08	3.35	3.39	4.52	-1.2%	-26.8%
0092-091 side	81.0	81.1	0.08	0.73	-	1.81	6.48	-	-
0092-071 side	706.4	704.7	0.71	0.99	-	15.47	18.76	-	-
002-006	66.1	66.0	0.73	0.84	3.03	3.11	3.64	-2.6%	-29.3%
071-092	228.7	228.3	0.74	0.78	3.35	3.39	3.69	-1.2%	-10.1%
071-004	71.1	70.9	0.66	0.69	2.82	2.86	3.47	-1.4%	-25.4%
071-000	0.0	0.0	0.66	0.99	2.80	2.89	4.23	-8.7%	-33.4%
002-070	203.9	203.9	0.74	0.71	3.12	3.13	7.89	-0.6%	-18.6%
004-011	71.1	70.8	0.68	0.63	1.85	1.88	1.93	-0.4%	-4.3%
004-012	69.0	69.1	0.68	0.68	1.79	1.87	2.87	-4.7%	-48.2%

IEEE benchmark [1] W. R. Knaeg and G. Stroh, "The critical analysis of IEEE test feeders," IEEE PES T&D, 2011 (submitted for review). DG not; Paper, reference [1], Table 12.

Rated currents agree with the paper within 2 %. Rmax: this study is within 2 % of the IEEE benchmark at 10 of 11 nodes (paper 4.7 %); the paper is 4 % to 151 % higher. Rmin: on the 4.0 kV lines: 0.64 to 1.07 kA here, 0.59 to 1.28 kA in the paper.

Say

Rated currents agree with the paper within 2 per cent. My fault levels match the IEEE benchmark within 2 per cent; the paper's fault levels are higher.

Understand

Three sources are compared: my study, the reference paper, and the IEEE benchmark. Load currents match the paper. Fault currents do not: mine are lower. To find out which is right, I compared both with the IEEE benchmark. Mine are within 2 per cent of it (green column). The paper's are much higher (red column). This is why some of my results differ from the paper.

नेपालीमा

तीन स्रोतको तुलना गरिएको छ: मेरो अध्ययन, reference paper र IEEE benchmark। Load current paper सँग मिल्छ। Fault current भने मिल्दैन: मेरो कम छ। कुन सही हो भनेर थाहा पाउन मैले दुवैलाई IEEE benchmark सँग दाँजें। मेरो मान benchmark को 2 प्रतिशतभित्र छ (हरियो स्तम्भ)। Paper का मान धेरै बढी छन् (रातो स्तम्भ)। मेरा केही नतिजा paper भन्दा फरक हुनुको कारण यही हो।

If asked

Why are your fault levels lower than the paper's? My model includes the substation transformer and matches the IEEE short-circuit benchmark within 2 per cent. Two of the paper's values are even higher than its own value

at the feeder head, which is not possible in a radial feeder without DG.

SLIDE 14

Chapter IV: Fuse Coefficients - Side by Side

Chapter IV: Fuse Coefficients – Side by Side

Fuse	IF (A)	r (mV/A)	r (delayed) (s)	b (this study)	b (paper, Table 3)	Difference
P032	3418	0.133	2.87	6.33	6.38	+0.05
P033	4102	0.11	2.008	6.33	6.35	+0.02
P034	15671	0.439	8.772	6.33	6.32	+0.01
P045	3418	0.133	2.87	6.32	6.42	-0.09
P046	3122	0.136	3.125	6.42	6.47	-0.05
F-DC	3949	0.106	2.13	6.32	6.34	-0.02
P071	3390	0.034	1.601	6.27	6.08	+0.19
P071-1	3390	0.034	1.601	6.30	5.83	+0.47
P092	3390	0.034	1.601	6.39	6.13	+0.26
P092-R	3151	0.037	1.829	6.27	6.14	+0.13
P072	3121	0.037	1.829	6.44	6.19	+0.25
P072-2	3001	0.049	2.572	6.31	5.97	+0.34
P084	1857	0.113	2.778	6.31	6.13	+0.20
P011	1857	0.113	2.778	6.32	6.19	+0.13
P032	1874	0.112	2.86	6.47	6.17	+0.30

Fuse line: $\log t = a \log I + b$, with $a = -1.8$. The coefficients agree within 0.33 for all 15 fuses.

Say

The fuse coefficients agree with the paper's Table three within 0.33.

Understand

The coefficient b fixes where each fuse's line sits on the graph. The two columns, mine and the paper's, are close for all 15 fuses. Small differences come from the lower fault currents. One sentence is enough here.

नेपालीमा

गुणाङ्क b ले हरेक fuse को रेखा ग्राफमा कहाँ बस्छ भन्ने तोक्छ। मेरो र paper को, दुवै स्तम्भ 15 वटै fuse मा नजिक छन्। सानो फरक कम fault current का कारण आएको हो। यहाँ एक वाक्य नै पुग्छ।

SLIDE 15

Chapter IV: DG Connected, Conventional R2

Chapter IV: DG Connected, Conventional R2

Metric	Value
Cells held	24 of 39
Cells lost	15
Lost at nodes	633, 645, 646, DL, 675

Example: three-phase fault near 632. R2 trips 0.121 s; fuse melts 0.039 s.

- The fuse carries grid + DG current and melts before the fast trip.
- For some LG faults the single-setting R2 does not pick up the reverse DG current.



Say

With the DG and the same settings, only 24 of 39 cells hold coordination. The fuse melts before the fast trip.

Understand

Now the DG is switched on and nothing else is changed. Green ticks are cells that still work; red crosses are cells where the fuse melts first. Fifteen cells are lost. They are near the DG and upstream of R2: nodes 633, 645, 646, the distributed load, and 675.

नेपालीमा

अब DG चालु गरिन्छ र अरू केही बदलिँदैन। हरियो चिह्न भएका cell अझै ठीक छन्; रातो क्रस भएका cell मा fuse पहिले पगल्छ। पन्ध्र वटा cell बिग्रन्छन्। ती DG नजिक र R2 भन्दा माथि छन्: node 633, 645, 646, distributed load र 675।

If asked

Why exactly does it fail? Two reasons. The fuse carries grid plus DG current, so it melts sooner. And R2 sees only the reverse DG current, which is small, so with its forward setting it is slow or does not pick up at all.

SLIDE 16

Chapter IV: Coordination with DG - Side by Side

Chapter IV: Coordination with DG – Side by Side

This study – coordination held in 24 of 39 cells

Fault	632	633	634	645	646	671	692	675	680	684	652	611	DL
LG	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LL	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LLG	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LLL	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Reference paper [1], Fig. 14 – coordination held in 30 of 39 cells

Fault	632	633	634	645	646	671	692	675	680	684	652	611	DL
LG	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LL	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LLG	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
LLL	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Agree in 29 of 39 cells (22 held in both, 7 lost in both). Differ in 10: 8 lost only here (4 of them at DL), 2 lost only in the paper (645). With the DSEDR: 18 of 39 here, 39 of 39 in the paper.

16

Say

I hold coordination in 24 cells, and the paper in 30. We agree in 29 of the 39 cells.

Understand

The same result, cell by cell, next to the paper's. Top grid is mine, bottom is the paper's. In 29 cells we give the same answer. In 8 cells I lose coordination and the paper holds it; in 2 it is the other way round. The biggest difference is the distributed-load node.

नेपालीमा

उही नतिजा, cell-cell गरेर, paper को नतिजासँगै राखिएको छ। माथिको जाली मेरो, तलको paper को हो। 29 वटा cell मा हामी दुवैको उत्तर एउटै छ। 8 वटा cell मा मेरोमा coordination बिग्रन्छ तर paper मा कायम रहन्छ; 2 वटामा यसको उल्टो छ। सबैभन्दा ठूलो फरक distributed load को node मा छ।

If asked

Why do you lose more cells than the paper? My fault levels follow the IEEE benchmark, and the paper does not publish R2's settings or all fuse sizes. With the same method but different currents and fuses, the cell-by-cell result cannot be identical.

SLIDE 17

Chapter IV: Effect of the DG Penetration Level - Side by Side

Chapter IV: Effect of the DG Penetration Level – Side by Side

DG (%)	Node 633 study	Node 633 paper	Node 671 study	Node 671 paper
0	+2.1	+20	+28.0	+36
10	-3.1	+23	+21.5	+22
25	-9.6	+14	+16.4	+12
37	-13.9	+6	+13.4	+6
50	-17.9	-4	+10.7	0
75	-24.2	-9	+7.1	-8
100	-29.2	-16	+4.7	-14

Based on CTI (ms) and penetration level (%). Paper values read from Fig. 7 of [1].

- This study, is done: CTI at 633 from +4 ms to -51 ms; at 671 from +403 ms to -75 ms.
- The fuse carries grid + DG current, the recloser only the grid share.



This study (left) and the reference paper (right)

Outer ring: node 671, inner ring: node 633.

Same trend in both: the CTI falls steadily as the DG penetration rises. Node 633 turns negative from 10 % here and from 50 % in the paper, because the starting margin here is only +4 ms.

17

Say

As the DG grows from 0 to 100 per cent, the CTI falls: at node 633 from plus 4 to minus 51 milliseconds. The paper shows the same trend. This is why an adaptive setting is needed.

Understand

This slide shows the problem growing. The DG size is increased in seven steps and the margin (CTI) is measured each time. A positive CTI means the recloser wins; negative means the fuse melts first. The bigger the DG, the smaller the CTI. So a setting that is right for one DG size is wrong for another: the protection must adapt.

नेपालीमा

यो slide ले समस्या बढ्दै गएको देखाउँछ। DG को आकार सात चरणमा बढाइन्छ र हरेक पटक margin (CTI) नापिन्छ। CTI धनात्मक हुनु भनेको recloser ले जित्नु हो; ऋणात्मक हुनु भनेको fuse पहिले पग्लनु हो। DG जति ठूलो, CTI त्यति सानो। त्यसैले एउटा आकारको DG का लागि ठीक भएको setting अर्को आकारका लागि गलत हुन्छ: सुरक्षा प्रणालीले आफूलाई अवस्था अनुसार ढाल्नुपर्छ।

If asked

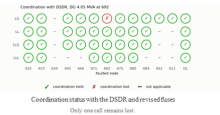
Why does the CTI fall? The fuse current rises with the DG, so the fuse melts faster. The recloser's current does not rise; it even falls a little. So the gap between them shrinks.

SLIDE 18

Chapter IV: With the DSDR

Chapter IV: With the DSDR

Scheme (DG connected)	Cells held
Conventional R2, original fuses	24
DSDR, original fuses	25
Conventional R2, revised fuses	31
DSDR, revised fuses	38 of 39



- The dual setting itself restores 7 cells: all faults at 633 and the LG faults at 645, 646 and DL.
- Still lost: LG fault at 692 – R2's delayed dial is already at its maximum.

18

Say

With the DSDR and revised fuses, 38 of 39 cells hold coordination. The dual setting alone restores seven cells. Only the LG fault at 692 stays lost, at R2's dial limit.

Understand

The main result. Only one red cross is left. The small table separates the two improvements: the dual setting alone gives 25, larger fuses alone give 31, and both together give 38. So neither is enough on its own. The remaining case cannot be fixed because R2's delayed dial is already at its maximum.

नेपालीमा

यो मुख्य नतिजा हो। एउटा मात्र रातो क्रस बाँकी छ। सानो तालिकाले दुई सुधारलाई छुट्याएर देखाउँछ: dual setting एकलैले 25, ठूला fuse एकलैले 31, र दुवै सँगै हुँदा 38। त्यसैले कुनै एउटा मात्र पर्याप्त छैन। बाँकी एउटा अवस्था सुधारन सकिँदैन, किनभने R2 को ढिलो dial पहिले नै अधिकतम मानमा छ।

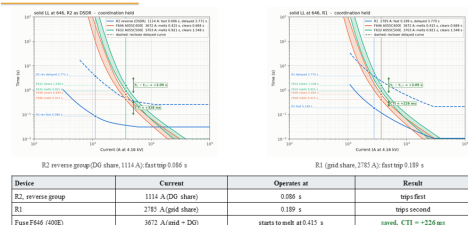
If asked

Then is the improvement from the DSDR or from the fuses? From both. With the revised fuses, a single setting gives 31 and the dual setting gives 38, so seven cells need the dual setting. With the old fuses the dual setting gives only 25. I call it necessary but not sufficient.

SLIDE 19

Chapter IV: TCC with the DSDR - Bolted LL Fault at 646

Chapter IV: TCC with the DSDR - Bolted LL Fault at 646



19

Say

For a line-to-line fault at 646, R2 trips in 0.086 seconds on its reverse group, and R1 in 0.189. The fuse would melt at 0.415, so it is saved.

Understand

One fault, drawn on the time-current graph. The fuse carries 3672 amperes, the grid and DG together. R2 sees only the DG part, 1114 amperes backwards, and trips first. R1 sees the grid part and trips second. Both are earlier than 0.415 seconds, when the fuse would begin to melt. The margin is 226 milliseconds.

नेपालीमा

एउटा fault लाई time-current ग्राफमा देखाइएको छ। Fuse मा 3672 ampere बग्छ, grid र DG दुवैको जोड। R2 ले DG को भाग मात्र, 1114 ampere उल्टो दिशामा देख्छ र पहिले trip गर्छ। R1 ले grid को भाग देख्छ र दोस्रोमा trip गर्छ। दुवै 0.415 सेकेन्डभन्दा अगाडि हुन्छन्, जुन बेला fuse पलन सुरु हुन्थ्यो। Margin 226 मिलिसेकेन्ड छ।

If asked

Why must both reclosers trip? The fault is fed from two sides. R2 cuts off the DG side and R1 cuts off the grid side. Only when both have opened does the current through the fuse stop.

SLIDE 20

Chapter IV: Operating Times with the DSDR - Side by Side

Chapter IV: Operating Times with the DSDR – Side by Side

This study (times in s)

Node	Fault	R1 fast	R1 delayed	R2 fast	R2 delayed	Fuse melts
632	LLL	0.081	1.627	0.021	1.418	-
633	LGL	0.106	2.112	0.080	2.051	0.149
643	LL	0.155	3.095	0.075	3.002	0.613
644	LL	0.189	3.770	0.086	3.771	0.415
65	LLL	0.106	2.129	0.048	1.213	0.121
671	LLL	0.134	2.677	0.034	1.600	-
692	LLL	0.134	2.677	0.034	1.600	0.029
675	LLL	0.119	3.122	0.039	1.842	0.065
684	LL	0.222	4.407	0.072	2.813	0.148
680	LLL	0.188	3.761	0.086	2.381	-
611	LGL	0.433	8.709	0.128	6.986	0.271
612	LGL	0.448	8.918	0.129	6.970	0.138

Reference paper [1], Table IV (times in s)

Node	Fault	R1 fast	R1 delayed	R2 fast	R2 delayed	Fuse melts
632	LLL	0.07	1.398	0.043	0.514	-
633	LGL	0.094	1.88	0.052	0.706	0.163
643	LL	0.105	2.095	0.087	1.659	0.276
644	LL	0.128	2.553	0.103	2.046	0.181
65	LLL	0.087	1.759	0.076	1.402	0.248
671	LLL	0.105	2.091	0.068	1.113	-
692	LLL	0.115	2.307	0.072	1.248	0.171
675	LLL	0.121	2.424	0.074	1.352	0.185
684	LL	0.125	2.507	0.075	1.394	0.205
680	LLL	0.137	2.747	0.079	1.555	-
611	LGL	0.326	6.520	0.146	4.383	0.417
612	LGL	0.676	13.529	0.143	9.37	0.239

Same behaviour in both: the fuse melts after the fast trip; R1 delayed = 20 × R1 fast (data 0.5 : 10). R1 is slower here in most rows because the fault levels follow the IEEE benchmark; R2's settings are not published in the paper.

20

Say

These are my operating times next to the paper's Table four. In both, the fuse melts after the fast trip in every row.

Understand

Left is my table, right is the paper's. Do not read the numbers. The point is the pattern: in every row of both tables, the fuse time is longer than the fast trip time. Also, R1's delayed time is exactly 20 times its fast time in both, which confirms the dials 0.5 and 10.

नेपालीमा

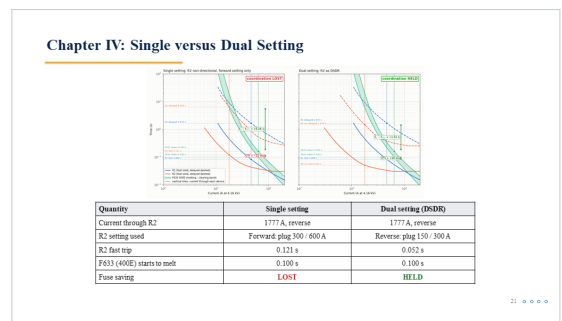
बायाँ मेरो तालिका, दायाँ paper को। सङ्ख्या नपढनुहोस्। मुख्य कुरा ढाँचा हो: दुवै तालिकाको हरेक पङ्क्तिमा fuse को समय छिटो trip को समयभन्दा लामो छ। साथै दुवैमा R1 को ढिलो समय छिटो समयको ठ्याक्कै 20 गुणा छ, जसले dial 0.5 र 10 हो भन्ने पुष्टि गर्छ।

If asked

Your times differ from the paper's. Why? The operating time depends on the fault current. My fault currents follow the IEEE benchmark and are lower, so R1 is slower in most rows. R2's settings are not published in the paper, so those can only be compared in trend.

SLIDE 21

Chapter IV: Single versus Dual Setting



Say

Same fault, same current. With the single setting, R2 trips in 0.121 seconds: coordination is lost. With the reverse group, 0.052 seconds: coordination is held.

Understand

The clearest proof that the dual setting matters. Everything is the same in both columns: the fault, the current, the fuse. Only R2's setting group is different. With the forward group R2 is too slow for this small reverse current. With the reverse group, which has a lower pickup, it is fast enough.

नेपालीमा

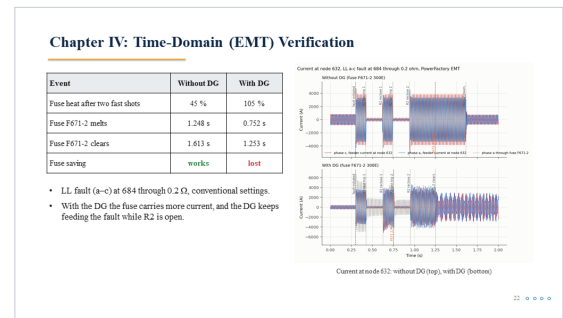
Dual setting महत्त्वपूर्ण छ भन्ने सबैभन्दा प्रस्ट प्रमाण यही हो। दुवै स्तम्भमा सबै कुरा उही छ: fault, current र fuse। फरक केवल R2 को setting समूह हो। Forward समूहमा R2 यो सानो reverse current का लागि धेरै ढिलो हुन्छ। कम pickup भएको reverse समूहमा भने पर्याप्त छिटो हुन्छ।

If asked

Why is the reverse group faster for the same current? Its pickup is lower: 150 amperes against 300. The same 1777 amperes is a larger multiple of the pickup, and a larger multiple gives a shorter time on an inverse curve.

SLIDE 22

Chapter IV: Time-Domain (EMT) Verification



Say

The EMT run confirms it in time. Without DG the fuse survives the two fast shots at 45 per cent of its heat. With DG it melts during the second shot.

Understand

This is a simulation of the actual current wave. The recloser trips, waits, recloses, and trips again: two fast shots. A fuse melts when it has collected enough heat. Without DG the two shots use 45 per cent of that heat, so the fuse survives. With DG the fuse current is higher, and the DG keeps feeding the fault even while R2 is open, so the fuse reaches 100 per cent at 0.75 seconds.

नेपालीमा

यो वास्तविक current को तरङ्गको simulation हो। Recloser ले trip गर्छ, पर्खन्छ, फेरि बन्द हुन्छ र फेरि trip गर्छ: दुई वटा छिटो shot। Fuse ले पर्याप्त ताप जम्मा गरेपछि पलन्छ। DG नहुँदा दुई shot ले त्यो तापको 45 प्रतिशत मात्र प्रयोग गर्छन्, त्यसैले fuse बन्छ। DG हुँदा fuse को current बढी हुन्छ, र R2 खुला हुँदा पनि DG ले fault लाई current दिइरहन्छ, त्यसैले fuse 0.75 सेकेन्डमा 100 प्रतिशत पुग्छ।

If asked

Why is the EMT run done with conventional settings? The fault at 684 is downstream of R2, where R2 uses its forward group in both schemes. The run shows the cause of the problem in time; it also shows the limit of the method for faults below R2.

SLIDE 23

Chapter IV: Summary and Comparison

Chapter IV: Summary and Comparison

Stage	This study (of 39)	Reference paper [1]
No DG, starting fuse sizes	35	—
No DG, after the fuse revision	39	—
DG connected, conventional R2	24	30
DG connected, original fuses, R2 as DSDR	25	—
DG connected, revised fuses, conventional R2	31	—
DG connected, revised fuses, R2 as DSDR	38	39

Same trend as the reference paper. The counts differ because the model is validated against the IEEE short-circuit benchmark (feeder head 4.75 kA) and uses real data ranges.

Say

In summary, coordination holds in 24 cells with a conventional recloser, 25 with the DSDR alone, 31 with the fuse revision alone, and 38 with both. The paper reports 30 and 39.

Understand

All the counts in one table. Read down the middle column: 35, 39, 24, 25, 31, 38. The right column gives the paper's two published counts, 30 and 39. The trend is the same in both: the DG reduces coordination, and the DSDR with the fuse revision restores it.

नेपालीमा

सबै गन्ती एउटै तालिकामा छन्। बीचको स्तम्भ माथिबाट तल पढ्नुहोस्: 35, 39, 24, 25, 31, 38। दायाँको स्तम्भमा paper मा प्रकाशित दुई गन्ती, 30 र 39 छन्। दुवैमा प्रवृत्ति उही छ: DG ले coordination घटाउँछ, र DSDR सँगै fuse को परिवर्तनले त्यसलाई फेरि कायम गराउँछ।

If asked

The paper gets 39 and you get 38. Is your result worse? It is one cell, the LG fault at 692. There the fuse must be large to stay selective with the fuse below it, and then it clears after R2's delayed trip. R2's delayed dial is already at its maximum, so a real CDG34 relay cannot do better.

SLIDE 24

Chapter V: Conclusion and Future Scope

Chapter V: Conclusion and Future Scope

Conclusion

- The DSDR method was implemented on the IEEE 13-node feeder in DigSILENT PowerFactory; the model meets the IEEE benchmark within about 2 %.
- Without DG, 35 of 39 cells with the starting fuses, 39 of 39 after the fuse revision.
- With the DG, a conventional R2 keeps only 24 of 39 cells: the fuse carries grid + DG current and melts before the fast trip.
- DSDR alone: 25 fuse revision alone: 31; DSDR with the fuse revision: 38 of 39 (zero-margin criterion).
- The DSDR is necessary but not sufficient: it needs the fuse revision; the LG fault at 692 is limited by R2's delayed dial.

Recommendation and Future Scope

- Extend the study to the IEEE 34-node-feeder and to inverter-based DG.
- Model an explicit directional element for R2, and change the setting group with the state of the DG.
- Use a recloser with a wider dial range to recover the remaining cell, validate with hardware-in-the-loop tests.

24 ○ ○ ○ ○

Say

To conclude: the DSDR with the fuse revision raises coordination from 24 to 38 of 39 cells. It is necessary but not sufficient, because it needs the fuse revision. Future work includes the 34-node feeder, a true directional element and inverter-based DG.

Understand

The conclusion repeats the story in five lines: the model is valid, it works without DG, the DG breaks it, the DSDR with larger fuses repairs it, and there is a limit. The future work lists what was not done: a larger feeder, a relay model that senses direction itself, and DG connected through inverters, which behaves differently in a fault.

नेपालीमा

निष्कर्षले पूरै कथालाई पाँच पङ्क्तिमा दोहोर्‍याउँछ: model ठीक छ, DG बिना यसले काम गर्छ, DG ले यसलाई बिगार्छ, DSDR र ठूला fuse ले सुधारछन्, र एउटा सीमा छ। Future work मा गर्न बाँकी काम छन्: ठूलो feeder, दिशा आफैं चिन्ने relay को model, र inverter मार्फत जोडिएको DG, जसले fault को बेला फरक व्यवहार गर्छ।

If asked

What is the main limitation of your work? The library relay is not directional, so the two groups are modelled as separate relay units and the direction is assigned from the fault location. Also, coordination is judged with a zero margin, and only one DG location is studied.

SLIDE 25

References

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- [7] W. H. Kersting and G. Sherk, "Short circuit analysis of IEEE test feeders," in *Proc. IEEE PES T&D*, 2012.
- [8] DigSILENT GmbH, *PowerFactory 2021 User Manual*, Garmisch, Germany, 2021.

25 ◀ ◻ ◻ ◻

Say

These are the main references.

Understand

Reference 1 is the method. References 5 and 7 are the test feeder and its short-circuit benchmark. Reference 4 is the relay manual for R1's curve.

नेपालीमा

Reference 1 विधिको स्रोत हो। Reference 5 र 7 test feeder र त्यसको short-circuit benchmark हुन्। Reference 4 R1 को curve का लागि प्रयोग गरिएको relay को manual हो।

SLIDE 26

Thank You

Thank You

Questions and Discussion

26 ◻ ◻ ◻

Say

Thank you. I am happy to take your questions.

Understand

Stop here and wait. If you do not understand a question, ask the examiner to repeat it.

नेपालीमा

यहाँ रोकिनुहोस् र पर्खनुहोस्। प्रश्न नबुझे परीक्षकलाई फेरि भन्न अनुरोध गर्नुहोस्।

4. Numbers to remember

Quantity	Value
Feeder voltage	4.16 kV
DG	4.05 MVA synchronous, at node 692
Cells studied	39 (12 locations, 4 fault types)
R1 pickup and dials	720 A; TDS 0.5 / 10
R2 forward plugs	300 / 600 A
R2 reverse plugs	150 / 300 A
Current through R2	470 A without DG; 252 A reversed with DG
Reverse pickup	$1.25 \times 252 = 315$ A
Fault level at feeder head	4.73 kA
No DG	35 of 39, then 39 of 39
DG, conventional R2	24 of 39
DSDR alone / fuses alone / both	25 / 31 / 38 of 39
Paper	30 and 39
Cell still lost	LG fault at 692
CTI at 633, DG 0 to 100 per cent	+4 ms to -51 ms
EMT: fuse heat after two fast shots	45 per cent without DG; melts at 0.75 s with DG